Sky View Factor & urban comfort analysis

What this tool produces · *Calculations and thermal mappings quantifying the proportion of sky visible from a specific point, proving how street canyons trap heat and dictating where to plant trees or install shade structures,.*

G1 · Proof of Impact

Dimensions

Environmental **Spatial**

aktan

With simplexity

Preparing the workshop · *Sky View Factor & urban comfort analysis*

G1 · Proof of Impact

Why this tool matters

Translating the harsh, bodily reality of a summer heatwave into objective geometry; proving mathematically that shade is not a mere aesthetic choice, but a vital necessity for human survival in the street.

Evidence we produce

Calculations and thermal mappings quantifying the proportion of sky visible from a specific point, proving how street canyons trap heat and dictating where to plant trees or install shade structures,.

Duration & Material

Duration · 2 to 4 days of field photography, 1 week of digital processing

Material · Fisheye lens cameras, SVF calculation software (RayMan), meteorological data

Reference case

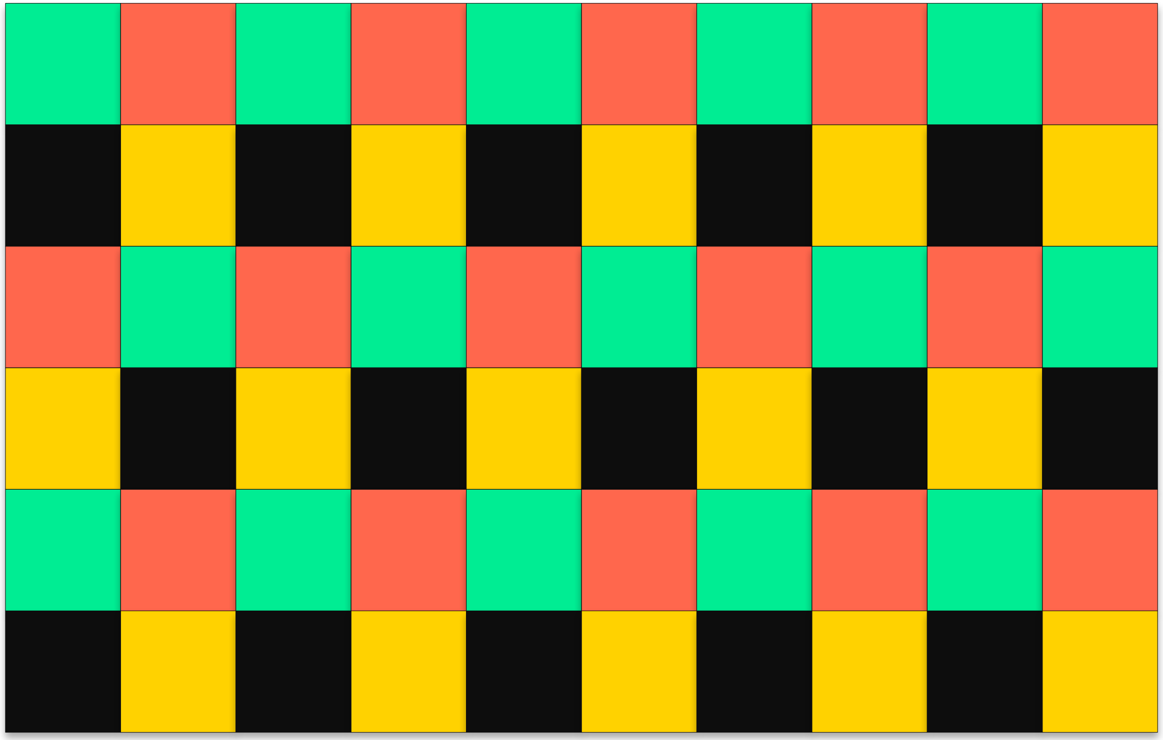
In a hot arid university campus plaza, researchers calculated the Sky View Factor using fisheye imagery and the RayMan model. By correlating the geometric openness of the plaza with microclimatic variables, they proved exactly how the lack of shading exacerbated pedestrian thermal discomfort, directly informing the architectural redesign of the campus to protect students from severe heat stress.

Suggested agenda · Brief (10 min) › Method walk-through (15 min) › Animation canvas (60–90 min) › Harvest (20 min)

SVF heatmap × comfort drivers

Duration
2 to 4 days of field photography, 1 week of digital pro...

Sky view factor heatmap



Comfort drivers

- **Air temperature**
- **Humidity**
- **Wind speed**
- **Solar radiation**
- **Surface materials**
- **Vegetation cover**

Harvesting our *learnings*

After running the canvas, capture three layers of insight.

What inspires me

Stand-out signals, surprising stories, ideas to remember

How to apply it here

Concrete adaptations to our context, our team, our constraints

What we need to be autonomous

Skills, allies, resources, decisions still to secure

Each participant fills a copy. Then debrief together to surface patterns and next steps.